Matrices of Maps

Chapter 2 built the theory of linear maps without ever choosing a basis, and Chapter 3 built the machinery of elimination for matrices. This chapter joins them. Fix a basis of the domain and one of the target, write each $T(v_j)$ in coordinates, and its coordinates become the columns of a matrix. Every linear map between finite-dimensional spaces becomes a matrix, composition becomes matrix multiplication — which is why the product is defined the way it is — and every question about a map becomes a question about an array of scalars that Chapter 3 knows how to reduce.

What we gain is computation; what we give up is uniqueness, because the matrix depends on the bases. So the chapter's real subject is the dependence itself. Changing basis conjugates the matrix, $\mathbf{A} \mapsto \mathbf{P}^{-1}\mathbf{AP}$, and two matrices represent the same operator exactly when they are **similar** — the question that the rest of the book spends five chapters answering. When both bases may be chosen freely the answer is immediate and disappointing in its simplicity: every matrix is equivalent to $\begin{pmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}$, and rank is the only invariant. The chapter closes with two constructions that the coordinate machinery makes routine: products and quotients of spaces with the isomorphism theorems, and projections and the trace, a number attached to an operator that no choice of basis can change.

What you need. From Chapter 2, all of it, and most of all the freedom to prescribe a linear map on a basis (Theorem 2.1.5), which is what makes the correspondence between maps and matrices a bijection: kernel and image with the injectivity test (Definition 2.2.1, Definition 2.2.2, Theorem 2.2.1, Theorem 2.2.2, Theorem 2.2.3, Theorem 2.2.4), rank, nullity, and Rank–Nullity with its counting corollaries (Definition 2.3.2, Definition 2.3.1, Theorem 2.3.1, Corollary 2.3.2), the algebra $\mathcal{L}(V)$ and polynomials of an operator (Definition 2.4.1, Theorem 2.4.1, Theorem 2.4.2, Definition 2.4.2), and isomorphisms — above all the coordinate isomorphism $v \mapsto [v]_{\mathcal{B}}$, which every proof here runs through (Corollary 2.5.5, Theorem 2.5.1, Theorem 2.5.4, Theorem 2.5.6, Theorem 2.5.7).

From Chapter 3, the matrix side of the same story: row reduction and the reduced form it produces (Definition 3.2.4, Theorem 3.2.3, Theorem 3.3.4), elementary matrices and the Invertible Matrix Theorem (Theorem 3.4.1, Theorem 3.4.4, Theorem 3.4.5), and the subspaces and numbers attached to a matrix, which §01 shows are the map's own (Definition 3.5.3, Definition 3.5.1, Definition 3.5.4, Theorem 3.5.4, Theorem 3.5.2, Theorem 3.5.5, Theorem 3.5.7, Theorem 3.5.9).

From Chapter 0, matrix arithmetic, since that is what a matrix of a map is made of: the product read three ways, its algebra, the transpose and similarity of matrices (Definition 0.9.9, Theorem 0.9.2, Theorem 0.9.4, Definition 0.9.10, Theorem 0.9.5, Example (Similarity)), the trace with its cyclic property (Theorem 0.9.6), and — for §04 only — equivalence classes and the check that a function on a quotient is well defined (Definition 0.5.2, Theorem 0.5.3). From Chapter 1, bases, coordinates, and dimension (Definition 1.5.1, Definition 1.6.1, Theorem 1.5.2, Theorem 1.6.4, Theorem 1.6.5), and direct sums with complements (Theorem 1.7.4, Theorem 1.7.6), which §05 reads through operators.

4.-1.1 Roadmap

- **The matrix of a linear map.** Coordinates turn maps into matrices, composition into matrix multiplication, and $\mathcal{L}(V, W)$ into $M_{m\times n}(F)$; rank and invertibility of a map are read off any matrix for it.
- **Change of basis and similarity.** How the matrix changes when the bases do, the change-

of-coordinates matrix, and similarity as the relation "same operator, different basis", with the invariants it preserves.

- **Rank, equivalence, and factorizations.** With both bases free, every matrix reduces to $\begin{pmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}$; rank factorizations, and the rank inequalities that follow from them.
- **Products and quotient spaces.** Building new spaces from old — the external product, and the quotient V/U whose vectors are cosets — with the isomorphism theorems that relate them.
- **Projections and the trace of an operator.** Direct sums seen through the idempotent operators they produce, and a trace that belongs to the operator rather than to any of its matrices.

4.-1.2 Named moves

- **Abstract space → write down a matrix.** Choose a basis that makes the map simple, compute with the matrix, and translate back through the coordinate isomorphism.
- **The change-of-basis square.** Do not recompute a matrix in a new basis: go around the square instead of across it, $[T]_{\mathcal{B}'}^{\mathcal{B}'} = \mathbf{P}^{-1}[T]_{\mathcal{B}}^{\mathcal{B}}\mathbf{P}$.
- **Free both bases.** When nothing forces the bases to match, choose them to suit the map; that is what turns an arbitrary matrix into a rank normal form.
- **Factor through the image.** Writing a rank r map as a surjection onto an r-dimensional space followed by an injection proves the rank inequalities in one move.
- **Check well-definedness on cosets.** Every operation on a quotient space, and every map out of one, begins by showing the answer does not depend on the representative chosen.
- **Recognize a direct sum by an idempotent.** $P^2 = P$ splits V as $\operatorname{im} P \oplus \ker P$, and conversely; a projection carries the complement, not just the image.